

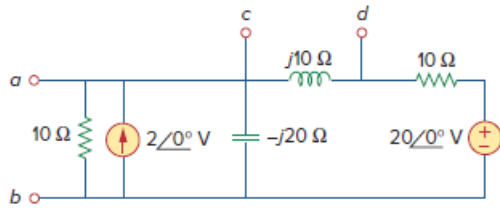
Problem

Find the Thevenin equivalent of the circuit in Fig. 10.103 as seen from:

(a) terminals a - b

(b) terminals c - d

Figure 10.103:



Step-by-step solution

Step 1 of 10

Refer to Figure 10.103 in the textbook.

(a)

Consider the circuit diagram to calculate the Thevenin impedance seen from a - b terminals.

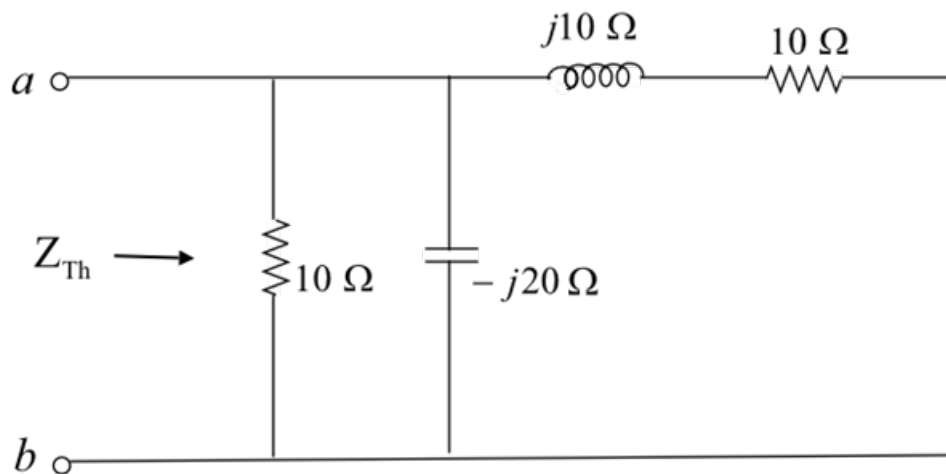


Figure 1

Step 2 of 10

In Figure 1, the $-j20\ \Omega$ reactance and $10\ \Omega$ resistors are connected in parallel. Therefore, the equivalent impedance is,

$$\begin{aligned} Z_1 &= (-j20) \parallel (10) \\ &= \frac{(-j20)(10)}{10 - j20} \\ &= (8 - j4)\ \Omega \end{aligned}$$

In Figure 1, the $j10\ \Omega$ reactance and $10\ \Omega$ resistors are connected in series. Therefore, the equivalent impedance is,

$$Z_2 = 10 + j10\ \Omega$$

Calculate the Thevenin impedance.

$$\begin{aligned} Z_{th} &= \frac{Z_1 Z_2}{Z_1 + Z_2} \\ &= \frac{(8 - j4)(10 + j10)}{8 - j4 + 10 + j10} \\ &= 6.67\ \Omega \end{aligned}$$

Therefore, the Thevenin's impedance, Z_{th} is $6.67\ \Omega$.

Step 3 of 10

Consider the circuit diagram to calculate the Thevenin voltage across a-b terminals.

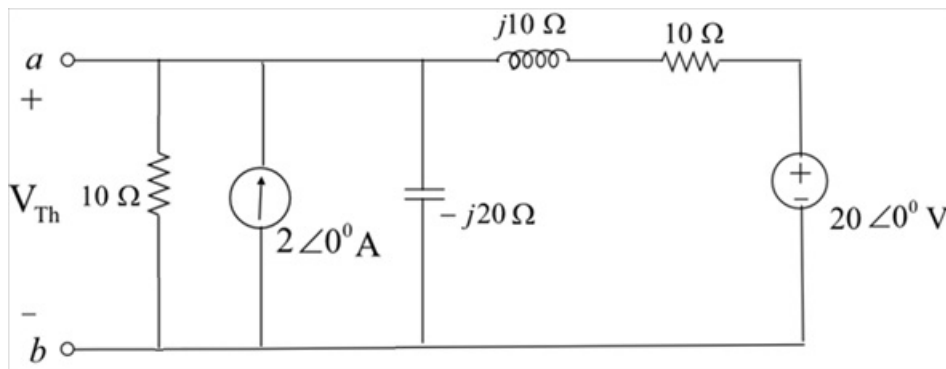


Figure 2

Step 4 of 10

Apply Kirchhoff's current law at node V_{th} .

$$\begin{aligned} \frac{V_{th}}{10} - 2 + \frac{V_{th}}{-j20} + \frac{V_{th} - 20}{10 + j10} &= 0 \\ \left(\frac{1}{10} + \frac{1}{-j20} + \frac{1}{10 + j10} \right) V_{th} &= 2 + \frac{20}{10 + j10} \\ (0.15) V_{th} &= 2 + 1 - j1 \\ V_{th} &= \frac{3 - j1}{0.15} \\ &= 20 - j6.67\ \text{V} \\ &= 21.08 \angle -18.43^\circ\ \text{V} \end{aligned}$$

Therefore, the Thevenin voltage, V_{th} is $21.08 \angle -18.43^\circ\ \text{V}$.

Step 5 of 10

(b)

Consider the circuit diagram to calculate the Thevenin impedance seen from c-d terminals.

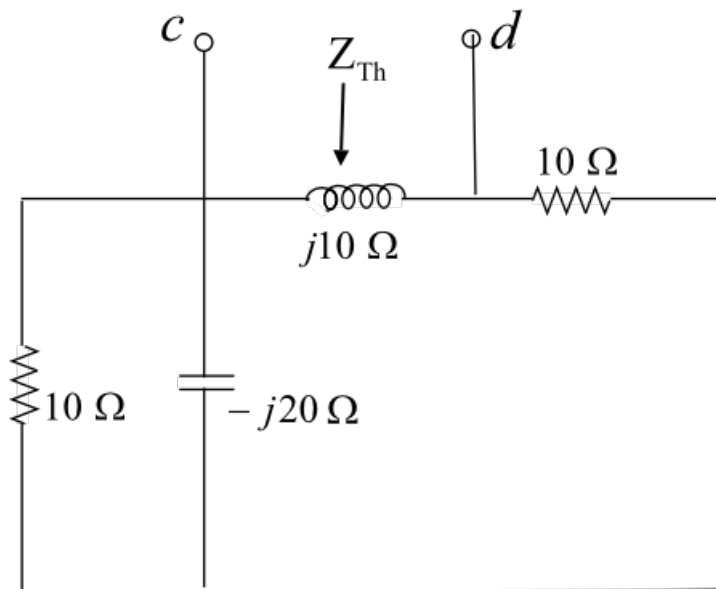


Figure 3

Step 6 of 10

In Figure 3, the $-j20\ \Omega$ reactance and $10\ \Omega$ resistors are connected in parallel. Therefore, the equivalent impedance is,

$$\begin{aligned} Z_1 &= (-j20) \parallel (10) \\ &= \frac{(-j20)(10)}{10 - j20} \\ &= (8 - j4)\ \Omega \end{aligned}$$

From Figure 3, calculate the Thevenin impedance.

$$\begin{aligned} Z_2 &= Z_1 + 10 \\ &= 8 - j4 + 10 \\ &= (18 - j4)\ \Omega \end{aligned}$$

The Thevenin's impedance is the parallel combination of $j10\ \Omega$ and Z_2 ; that is

$$\begin{aligned} Z_{th} &= (8 - j4 + 10) \parallel (j10) \\ &= \frac{(18 - j4)(j10)}{18 - j4 + j10} \\ &= 5 + j8.33\ \Omega \end{aligned}$$

Therefore, the Thevenin's impedance, Z_{Th} is $5 + j8.33\ \Omega$.

Step 7 of 10

Consider the circuit diagram to calculate the Thevenin voltage across c-d terminals.

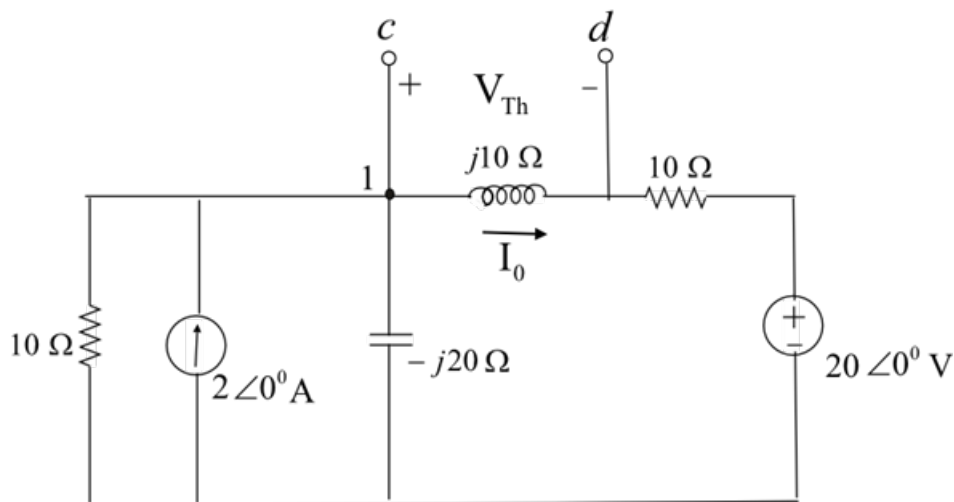


Figure 4

Step 8 of 10

Apply Kirchhoff's current law at node 1.

$$\begin{aligned}\frac{V_1}{10} - 2 + \frac{V_1}{-j20} + \frac{V_1 - 20}{10 + j10} &= 0 \\ \left(\frac{1}{10} + \frac{1}{-j20} + \frac{1}{10 + j10} \right) V_1 &= 2 + \frac{20}{10 + j10} \\ (0.15) V_1 &= 2 + 1 - j1 \\ V_1 &= \frac{3 - j1}{0.15} \\ &= 20 - j6.67 \text{ V} \\ &= 21.08 \angle -18.43^\circ \text{ V}\end{aligned}$$

Step 9 of 10

Calculate the current, I_0 .

$$\begin{aligned}I_0 &= \frac{V_1 - 20}{10 + j10} \\ &= \frac{20 - j6.67 - 20}{10 + j10} \\ &= \frac{6.67 \angle -90^\circ}{14.14 \angle 45^\circ} \\ &= 0.472 \angle -135^\circ \text{ A}\end{aligned}$$

Step 10 of 10

Calculate the Thevenin voltage.

$$\begin{aligned}V_{Th} &= (j10) I_0 \\ &= (10 \angle 90^\circ)(0.472 \angle -135^\circ) \\ &= 4.72 \angle -45^\circ \text{ V}\end{aligned}$$

Therefore, the Thevenin's voltage, V_{Th} is $4.72 \angle -45^\circ \text{ V}$.